

Skyler (Zican) Zhou

Email: skylerzzc@gmail.com | **Github:** <https://github.com/skylerzhou> | **Website:** <https://skylerzhou.github.io/>

EDUCATION

- University of Pennsylvania** | MSE in Bioengineering (3.82 / 4.00) Philadelphia, PA | May 2025
- Coursework: Brain-Computer Interface, Theoretical Neuroscience, Probability, Statistics for Data Science, Nonparametric Methods and Log-linear Models, Big Data Analytics (Python, PyTorch), Database & Information Systems (SQL)
- St. Lawrence University** | BS in Biology; Minor in Mathematics (magna cum laude) Canton, NY | May 2021

WORK EXPERIENCE

- Graduate Researcher** | Center for Neuroengineering and Therapeutics at UPenn Philadelphia, PA | June 2024 -
- Skills: ECoG Data Cleaning & Artifact Detection, Automated Pipeline Development, Statistical Analysis & Visualization (MATLAB, Python, Shell Script, R), Machine Learning (scikit-learn, PyTorch), Certificates in HIPAA and CITI Good Clinical Practice
- Built and validated scalable data processing pipelines for ECoG data, improving CCEPs (Cortico-cortical evoked potentials) detection sensitivity 4× while maintaining high precision (94% TPR). Ensured pipeline robustness and reproducibility across different research groups.
 - Developed statistical models (GLMM, Wilcoxon tests) to characterize seizure propagation across a heterogeneous patient cohort. Identified spatial and connectivity factors related to seizure spread. Leading manuscript preparation as first author with a cross-disciplinary team of clinicians, engineers, and scientists.
- Clinical Research Coordinator** | Pennsieve Data Management Platform Philadelphia, PA | July 2025 -
- Skills: Clinical Neuro Data Collection and Mangement (BIDS, CDEs), Application Development (Python, Shell Scripting, Docker), Backend Support for Platform Maintenance (Go, SQL), Coordination and Communication
- Coordinated between 5 institutes to develop and deploy Python-based iEEG/MRI data processing utilities (de-identification, BIDS/CDE-compliant metadata creation, migration, validation) on 300+ subjects' data.
 - Designed and built an interactive data dashboard for Epilepsy.Science (12+ TB multi-modal neuro data platform), enabling clinicians and researchers to effectively explore cohort demographics, modality availability, and pre/post-implant surgical outcomes.
 - Designed and integrated modality-specific visualization scripts into Pennsieve's AI Agent, reducing redundant LLM inference costs for recurring queries of EEG, MRI, and clinical data summaries.
- Research Assistant** | Computational Omics Lab at SUSTech Shenzhen, China | Dec 2021 - May 2023
- Skills: Data Collection from Public Datasets, Linux-based Large-scale Data Processing, Data Visualization (Python, R, Shell Script)
- Engineered modular data processing pipelines for genotype-phenotype association analysis on UK Biobank cohorts (~50k samples), emphasizing reproducibility and workflow scalability over repeated analyses. This work led to a poster presentation at the academic conference at the SUSTech School of Medicine.

SELECTED PROJECTS

- NeuroGrip;** Python, PyTorch Philadelphia, PA
- Designed time-series neural decoding models to forecast finger flexion from human ECoG recordings, applying frequency-domain feature extraction and predictive modeling to improve performance. The model achieved a correlation of 0.42 between predicted and actual movements, exceeding the performance reported in the prior literature.
- ParkDesktop;** Python, SQL, MYSQL, Node.js, React, Google Maps API, AI Agent Philadelphia, PA
- Built a full-stack data application integrating Python, SQL, and a React frontend to process, aggregate, and visualize large-scale NYC transportation safety data via an interactive map interface.

SKILLS

- Programming:** Python (NumPy, Pandas, scikit-learn, PyTorch), MATLAB, SQL, R, Shell Scripting, Go
- Signal Processing:** EEG/iEEG Signal Processing, Filtering, Fourier Transform, Detrending, Artifact Detection, Feature Extraction
- Modeling and Evaluation:** Predictive Modeling, Performance Tuning, Statistical Analysis, Visualization
- Software Development:** Pipelines, Docker, Linux, SSH, Git, GitHub Copilot, Claude Coworker, Codex